

Class-conditional sourcing

the two classes are obtained by different procedures

counterfeit = screen capture,
authentic = downloaded photo

Two acquisition pipelines

format, encoder, resolution,
lighting and backdrop differ
by class

PNG vs JPEG; 405 vs 223 px;
339 vs 6 kB

Acquisition predicts the label

$I(Y; A) > 0$; complete
when $H(Y | A) = 0$

format alone: 510/510;
header model: 1.000

detectable here,
before any model is trained

Every partition inherits it

stratified, grouped and
cross-validated alike

9/480 groups leak; removing
them moves accuracy 0.3 pt

High in-distribution accuracy

which cannot separate
packaging semantics from
provenance

0.919 on the leakage-free
test partition

A new acquisition pipeline

evaluation on images the
authors did not collect

150 external authentic
photographs (Split C)

Failure

the shortcut the model
relies on is absent

3.3% correct